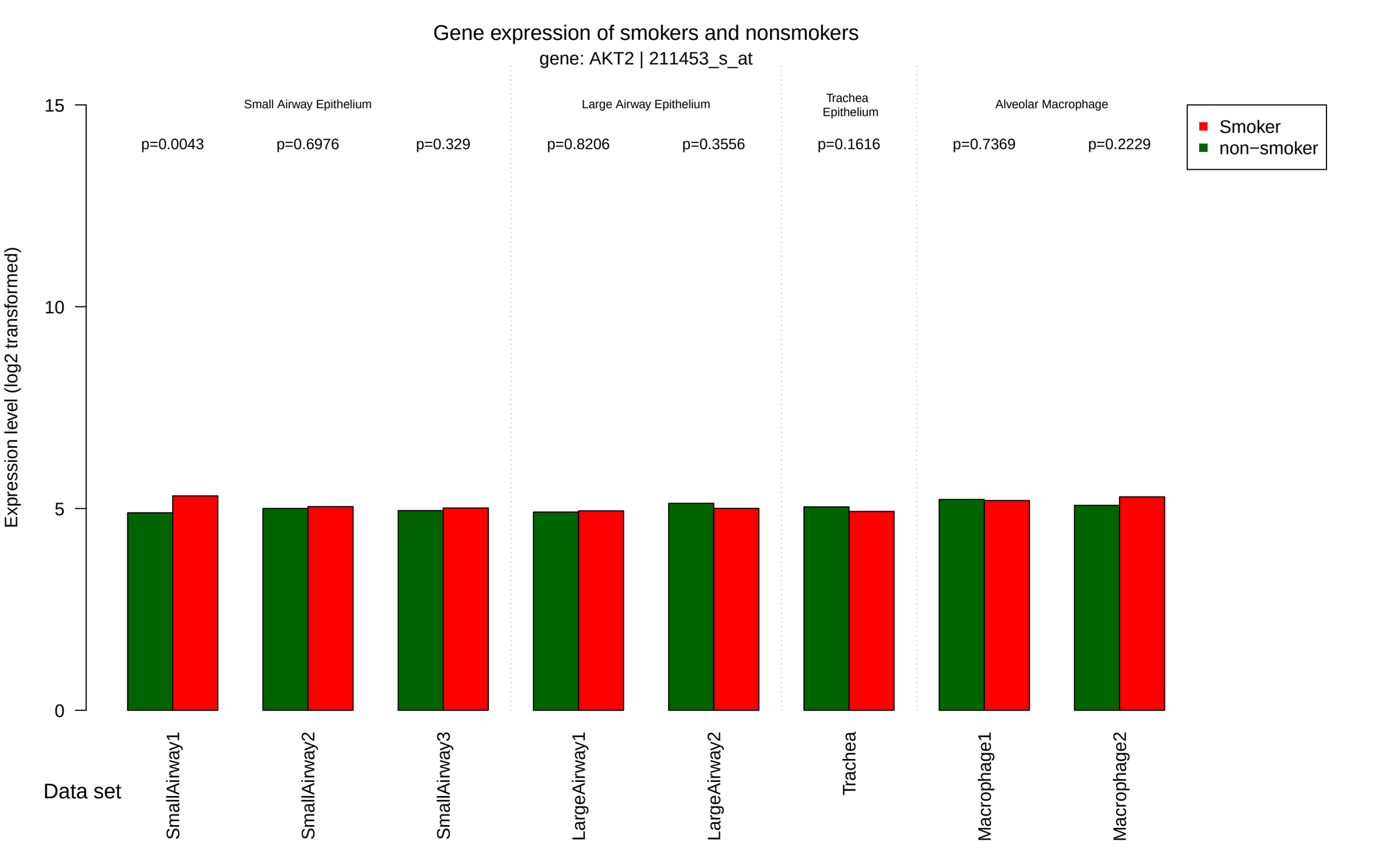
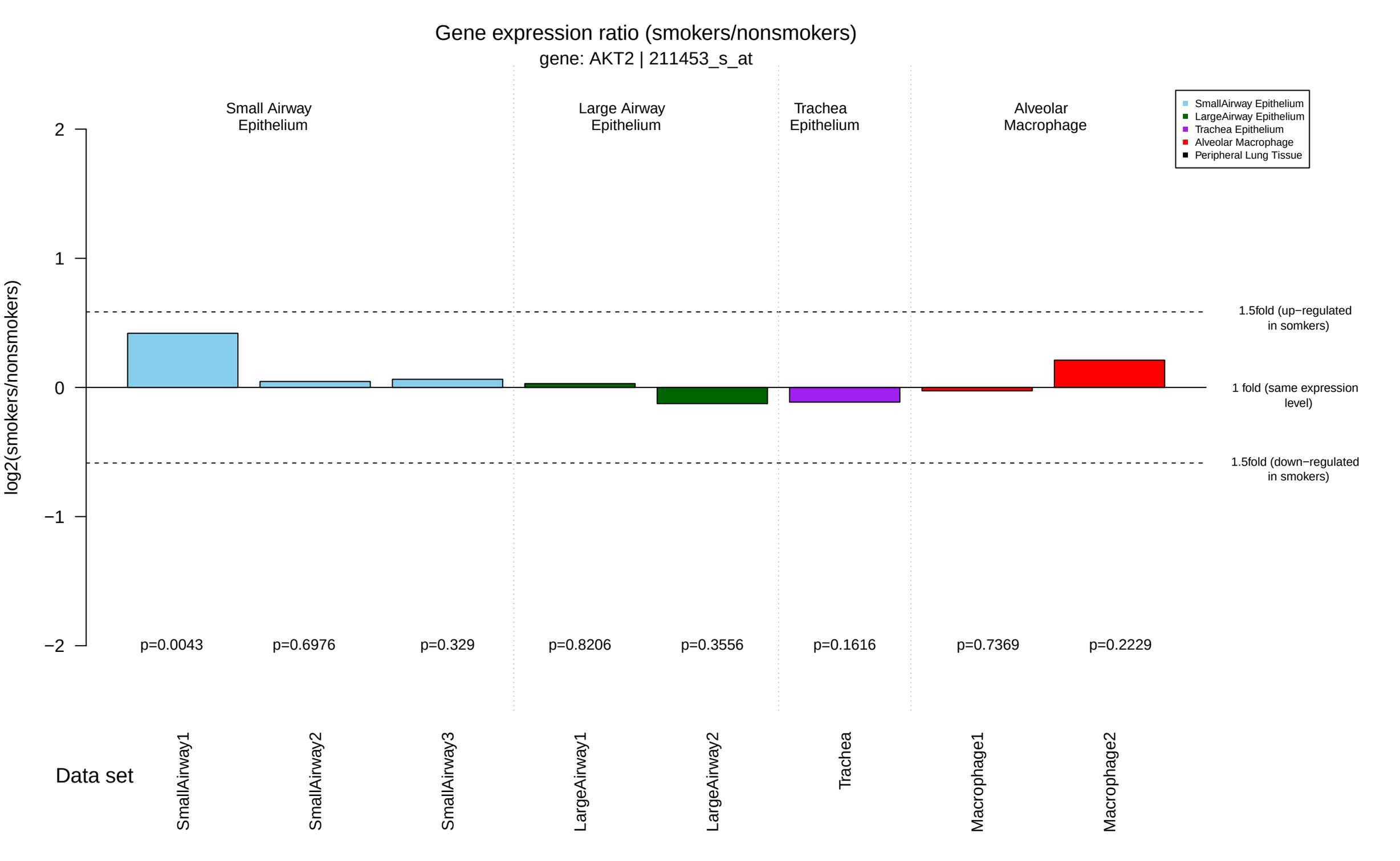
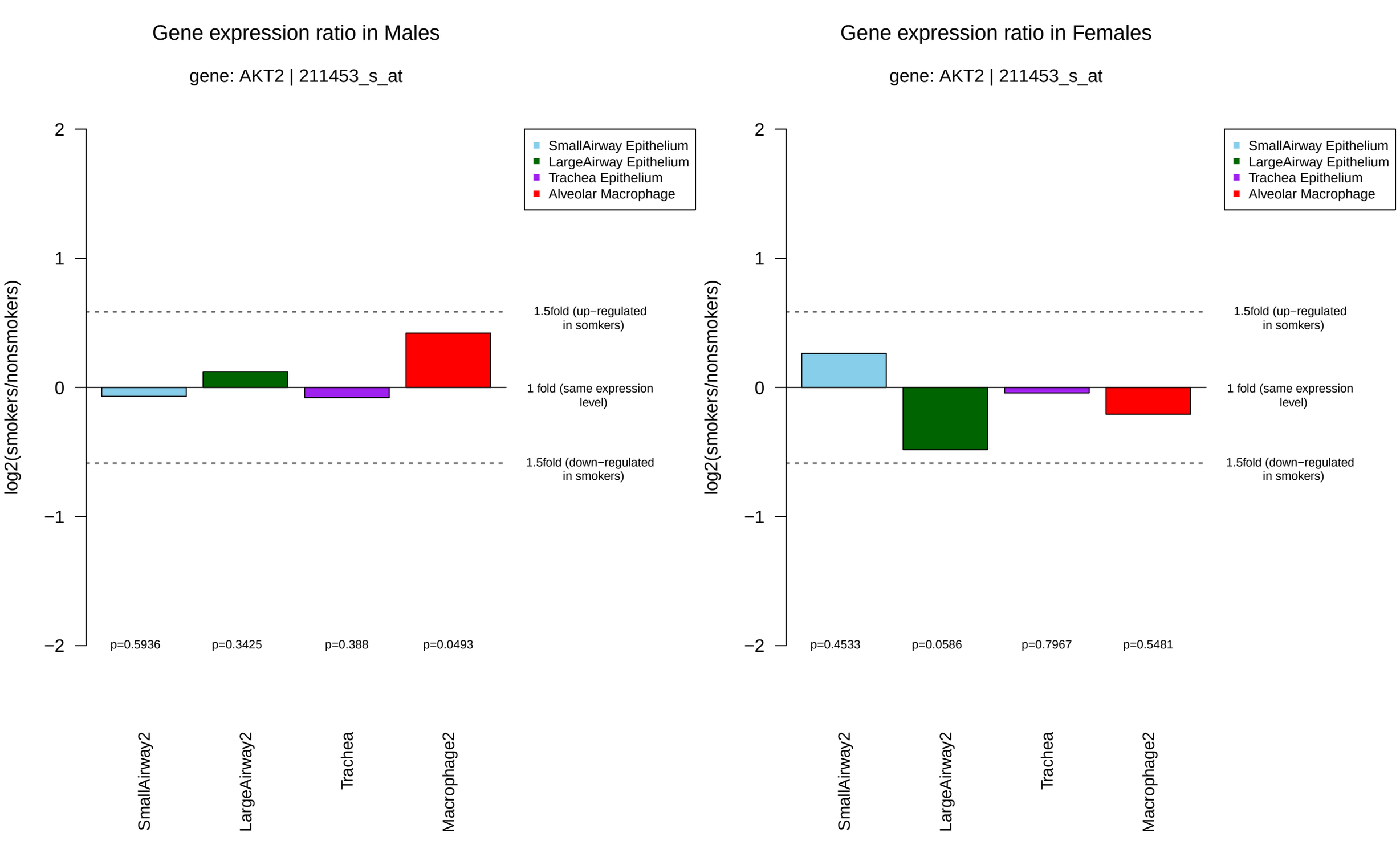
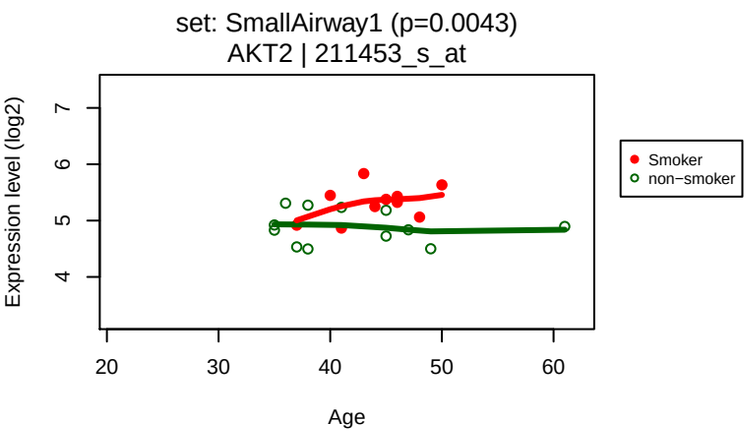
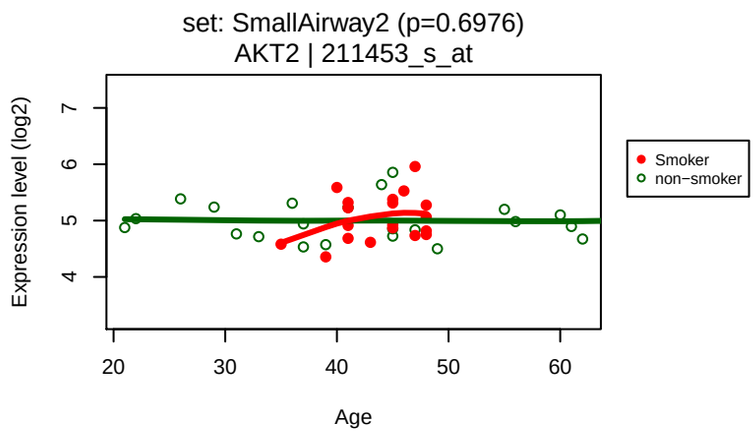
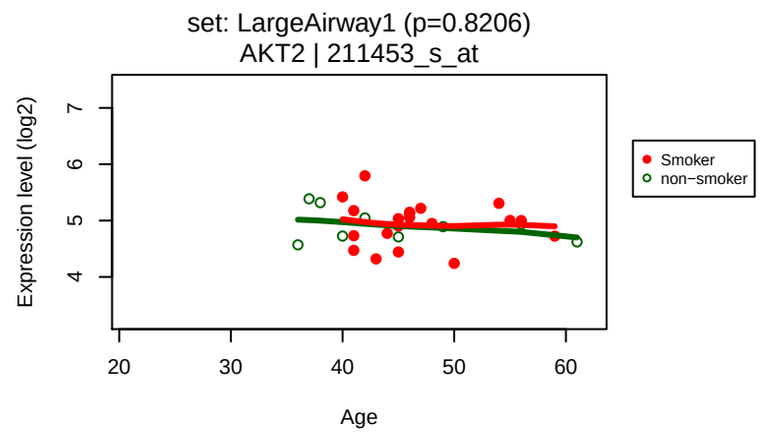

Gene expression of smokers and nonsmokers
gene: AKT2 | 211453_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0043
p=0.6976
p=0.329
p=0.8206
p=0.3556
p=0.1616
p=0.7369
p=0.2229
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

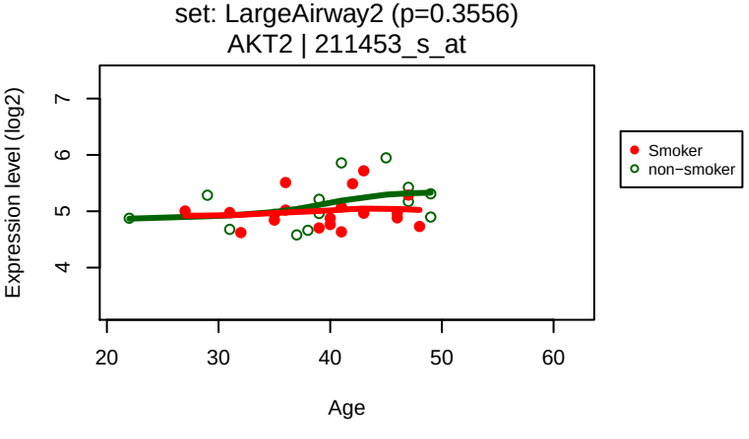

Gene expression ratio (smokers/nonsmokers)
gene: AKT2 | 211453_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0043
p=0.6976
p=0.329
p=0.8206
p=0.3556
p=0.1616
p=0.7369
p=0.2229
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

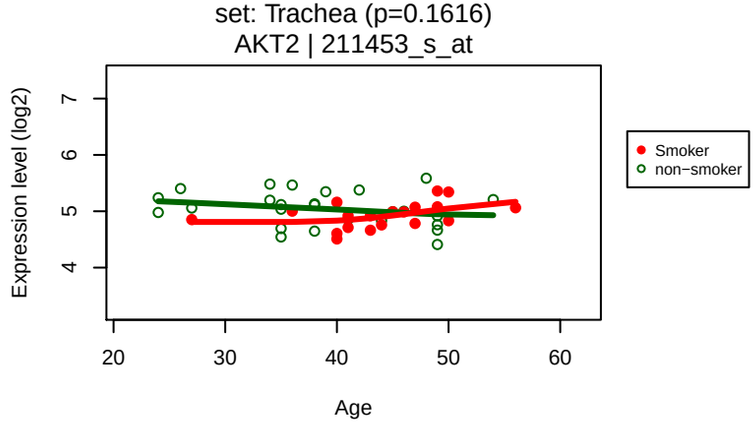

Gene expression ratio in Males
gene: AKT2 | 211453_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5936
p=0.3425
p=0.388
p=0.0493
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: AKT2 | 211453_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4533
p=0.0586
p=0.7967
p=0.5481
SmallAirway2
LargeAirway2
Trachea
Macrophage2

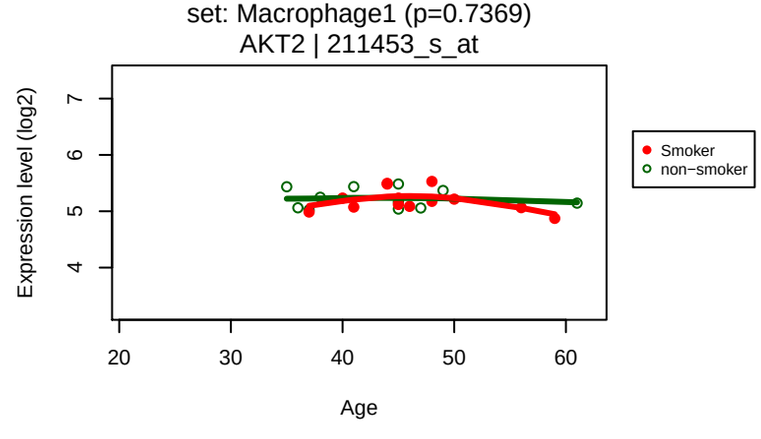

set: SmallAirway1 (p=0.0043)
AKT2 | 211453_s_at
Expression level (log2)
Age
Smoker
non−smoker

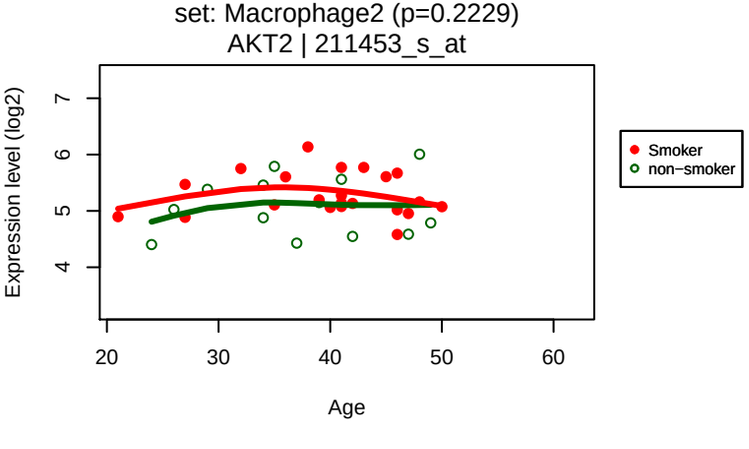

set: SmallAirway2 (p=0.6976)
AKT2 | 211453_s_at
Expression level (log2)
Age
Smoker
non−smoker

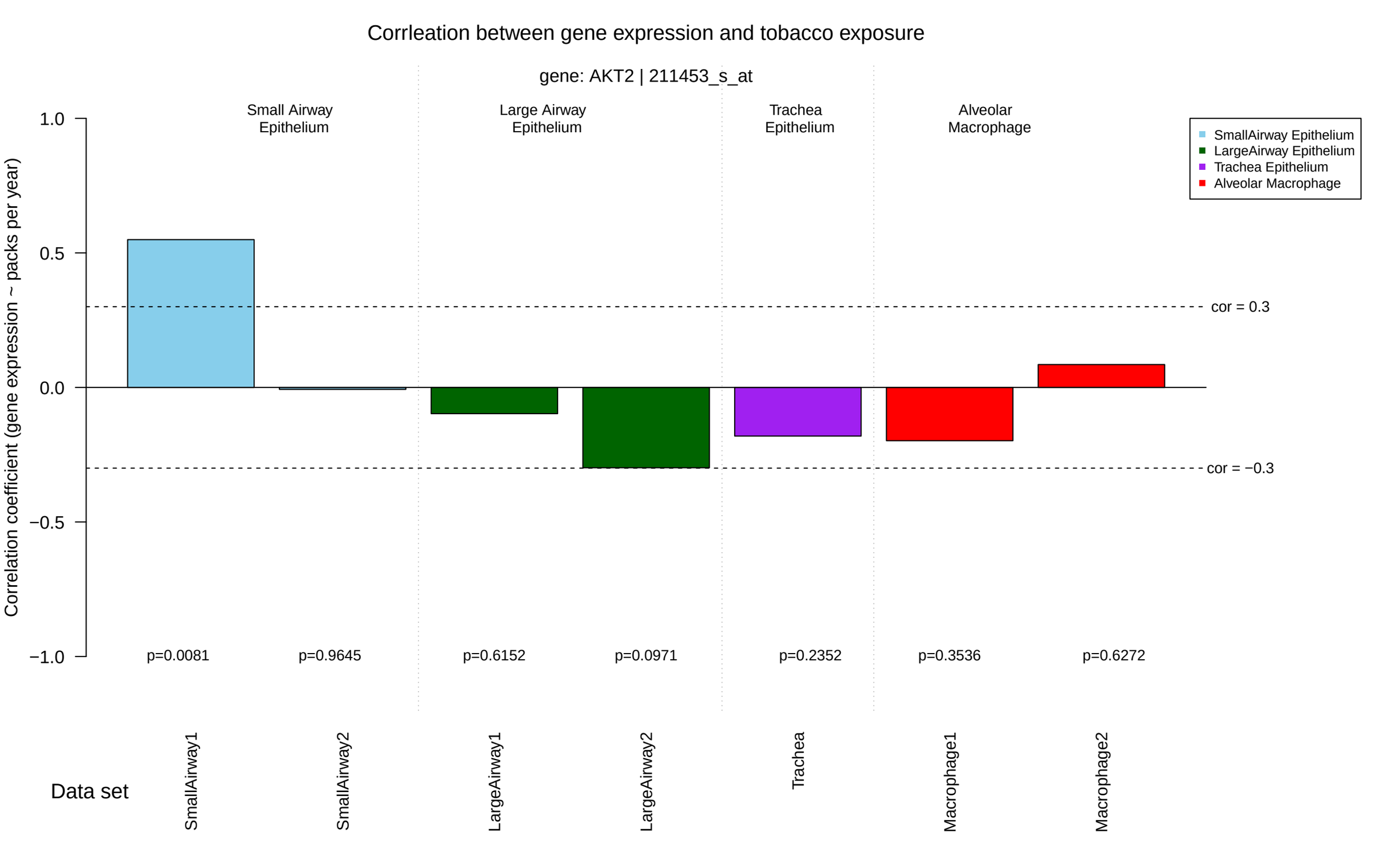

set: LargeAirway1 (p=0.8206)
AKT2 | 211453_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3556)
AKT2 | 211453_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1616)
AKT2 | 211453_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7369)
AKT2 | 211453_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2229)
AKT2 | 211453_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: AKT2 | 211453_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.0081
p=0.9645
p=0.6152
p=0.0971
p=0.2352
p=0.3536
p=0.6272
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2